

Preprocessing, Regularisation, and Generalisation in Epileptic Seizure Prediction: A Comparative Machine Learning Study

Abstract—Epileptic seizure prediction is a critical clinical problem where machine learning can deliver significant value. However, the performance of predictive models is heavily influenced by preprocessing choices, regularisation strategies, and class imbalance handling techniques. This study investigates how these factors affect generalisation across three distinct EEG-based datasets: the Bonn University EEG Signal dataset, a 19-channel ADHD EEG dataset, and the UCI Epileptic Seizure Recognition dataset. Using logistic regression as a controlled baseline model, we systematically compare two preprocessing pipelines, three regularisation penalties (L1, L2, and Elastic Net), and four imbalance handling strategies (no resampling, SMOTE, random undersampling, and class weighting). Our results demonstrate that the ordering of preprocessing steps materially affects model performance, L1 regularisation provides the best generalisation on high-dimensional data, and class weighting combined with Elastic Net regularisation consistently yields the strongest precision–recall tradeoff under class imbalance. These findings provide practical guidelines for clinical EEG classification tasks.

Index Terms—Epileptic seizure prediction, EEG classification, logistic regression, regularisation, class imbalance, SMOTE, preprocessing pipelines, L1/L2/Elastic Net

I. INTRODUCTION

Epilepsy affects approximately 50 million people worldwide, making it one of the most prevalent neurological disorders [1]. Accurate and timely seizure prediction from electroencephalography (EEG) signals can substantially improve patient quality of life by enabling pre-emptive intervention and reducing the risk of injury. Machine learning approaches to

seizure prediction have gained considerable traction over the past decade, yet several fundamental methodological questions remain underexplored in the literature.

First, EEG datasets are characterised by severe class imbalance: seizure episodes constitute only a small fraction of total recording time. Second, preprocessing choices — including normalisation, noise removal, feature extraction, and dimensionality reduction — are often applied in ad hoc orderings without systematic justification. Third, regularisation strategy selection is rarely benchmarked across multiple neurological datasets simultaneously.

This study addresses these gaps by conducting a controlled, reproducible comparison across three datasets with a logistic regression baseline. Although logistic regression is relatively simple, it provides a transparent platform to isolate the individual contributions of preprocessing, regularisation, and imbalance handling. Our contributions are:

- A systematic comparison of two preprocessing pipelines differing in step ordering, applied identically to all three datasets.
- A regularisation study comparing L1, L2, and Elastic Net penalties across a range of regularisation strengths, with explicit analysis of sparsity.
- An evaluation of four class imbalance strategies and their interaction with regularisation choice.

- Empirical answers to four critical comparative questions regarding generalisation in seizure prediction tasks.

II. DATASET COLLECTION AND JUSTIFICATION

Three publicly available EEG datasets were selected to provide diversity in size, class balance, and feature characteristics.

A. Dataset 1: EEG Signal — Bonn University

The Bonn University EEG dataset contains approximately 2 million single-channel raw signal samples recorded from both healthy subjects and epileptic patients across five labelled states: A (healthy, eyes open), B (healthy, eyes closed), C (interictal, opposite hemisphere), D (interictal, seizure focus zone), and E (ictal/seizure). For binary classification, class E is treated as the positive (seizure) class. Because the dataset contains a single raw signal column, statistical features are extracted using a sliding window approach (window size $w = 50$ samples), yielding nine features per window: mean, standard deviation, minimum, maximum, range, skewness, kurtosis, signal energy, and zero-crossing count.

Justification: This dataset represents the raw time-series paradigm, requiring explicit feature engineering. Its large size and moderate class imbalance ($\approx 4:1$) make it representative of real-world ambulatory monitoring conditions.

B. Dataset 2: ADHD EEG — 19-Channel Multi-Electrode

The ADHD EEG dataset contains recordings from 19 standard electrode positions (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T7, T8, P7, P8, Fz, Cz, Pz) from two subject groups: ADHD and neurotypical controls. The binary classification target is ADHD (1) vs. Control (0).

Justification: This dataset provides pre-extracted multi-channel features, near-balanced class distribution, and a neurological (non-epileptic) classification task that tests cross-domain generalisation of the pipeline.

C. Dataset 3: Epileptic Seizure Recognition — UCI

The UCI Epileptic Seizure Recognition dataset contains 11,500 one-second EEG segments represented by 178 extracted statistical features each. The original five-class target ($y \in \{1, 2, 3, 4, 5\}$) is binarised to seizure ($y = 1$) vs. non-seizure ($y \in \{2, 3, 4, 5\}$), yielding a 4:1 imbalance ratio.

Justification: This dataset is the most widely benchmarked in the seizure detection literature, providing a reference point for comparison. Its high-dimensional feature space (178 features) makes it particularly informative for evaluating feature selection and regularisation-induced sparsity.

D. Dataset Summary

TABLE I: Dataset Characteristics

Dataset	Rows (used)	Raw Features	Feature Type	Imbalance
DS1: EEG Signal (Bonn)	80,000	1 (raw signal)	Time-series	$\approx 4:1$
DS2: ADHD EEG	80,000	19	Extracted	$\approx 1:1$
DS3: UCI Seizure	11,500	178	Extracted	$\approx 4:1$

III. PREPROCESSING PIPELINES

Two distinct preprocessing pipelines were designed to investigate the effect of step ordering on model performance. All preprocessing transformations are fit on the training set only and applied to validation and test sets using the stored parameters, preventing data leakage.

A. Pipeline A: Normalisation $\rightarrow$ Noise Removal $\rightarrow$ Feature Selection

- 1) **Min-Max Normalisation:** Each feature x_j is scaled to $[0, 1]$:

$$x'_j = \frac{x_j - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (1)$$

- 2) **IQR Clipping (Noise Removal):** Outliers are clipped to the $[Q_1 - 1.5 \cdot \text{IQR}, Q_3 + 1.5 \cdot \text{IQR}]$ interval per feature, suppressing the influence of physiological artefacts and measurement noise.

- 3) **Feature Selection (SelectKBest)**: Features are ranked by their ANOVA F-statistic with respect to the class label, retaining the most informative subset.

B. Pipeline B: Scaling $\rightarrow$ PCA

- 1) **Standard Scaling**: Each feature is zero-centred and unit-variance normalised:

$$x'_j = \frac{x_j - \mu_j}{\sigma_j} \quad (2)$$

- 2) **Principal Component Analysis (PCA)**: The dimensionality is reduced to the minimum number of components that retain 95% of the explained variance:

$$\sum_{i=1}^k \lambda_i / \sum_{i=1}^n \lambda_i \geq 0.95 \quad (3)$$

where λ_i are the eigenvalues of the feature covariance matrix.

C. Research Insight: Ordering Matters

A key hypothesis of this study is that step ordering is not arbitrary. Applying PCA prior to standardisation distorts variance estimation because features with larger absolute magnitudes dominate the principal components. Concretely, if raw features have heterogeneous scales, an unscaled covariance matrix will inflate the importance of high-magnitude features, causing PCA to discard genuinely informative low-magnitude dimensions. Pipeline A mitigates this by normalising first; Pipeline B explicitly enforces standardisation before PCA. A third ordering (Scale $\rightarrow$ PCA $\rightarrow$ Select) is also tested in Section VIII.

IV. BASELINE MODEL: LOGISTIC REGRESSION

Logistic regression is selected as the baseline model for its interpretability, efficient training, and the direct interpretability of its regularisation parameters. The model estimates the posterior probability of the seizure class:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta^T \mathbf{x})}} \quad (4)$$

Parameters β_0 and β are estimated by minimising the binary cross-entropy loss via the L-BFGS solver

(or `liblinear` for L1 variants). All models are trained for up to 1000 iterations with convergence tolerance 10^{-4} .

A. Evaluation Metrics

Three metrics are reported throughout this study:

- **Accuracy**: Overall fraction of correct predictions. Less informative under class imbalance.
- **F1-Score**: Harmonic mean of precision and recall, suitable for imbalanced binary tasks:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

- **PR-AUC** (Area Under the Precision-Recall Curve): The primary metric for imbalanced datasets, measuring discrimination ability across all classification thresholds.

B. Baseline Results

TABLE II: Baseline Logistic Regression — Test Set Performance

Dataset	Pipeline	Accuracy	F1	PR-AUC
DS1	A	NaN	NaN	NaN
	B	NaN	NaN	NaN
DS2	A	0.5	0	0.5
	B	0.4986	0.5019	0.4840
DS3	A	0.8	0	0.4164
	B	0.8064	0.0618	0.4505

Note: Skipping DS1 with both Pipelines: Training data has only one class (0). Model cannot be trained.

V. OVERFITTING AND UNDERFITTING ANALYSIS

To demonstrate the bias-variance tradeoff, three configurations of logistic regression are deliberately constructed:

- **Underfitting** ($C = 0.0001$, 2 features): Extremely strong regularisation drives most coefficients to near-zero, severely limiting model capacity. The model exhibits high bias and low variance — both training and validation F1 are low and converge quickly with additional data.
- **Baseline** ($C = 1.0$, all features): Default regularisation provides a balance between bias and variance, producing a generalising model.

- **Overfitting** ($C = 10000$, all features): Near-zero regularisation allows the model to fit training noise. Training F1 approaches 1.0 while validation F1 lags significantly, with the gap persisting even as training set size increases.

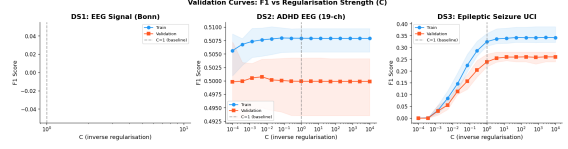


Fig. 2: Validation curves: F1-score vs. regularisation strength C for each dataset. The vertical dashed line marks $C = 1$ (baseline).

A. Learning Curves

Learning curves plot training and validation F1-score as a function of training set size, using 5-fold stratified cross-validation. The characteristic signatures of underfitting and overfitting are evident across all three datasets.

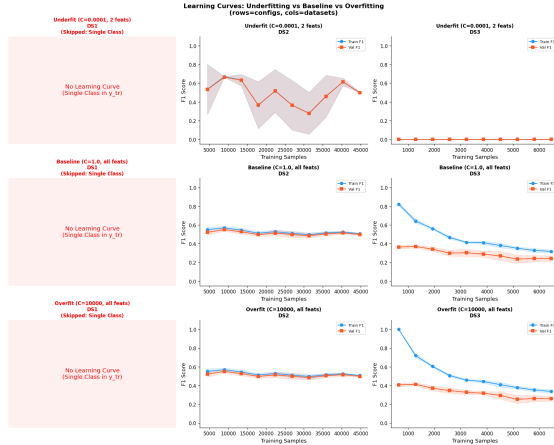


Fig. 1: Learning curves (Train F1 vs. Val F1) for underfitting, baseline, and overfitting configurations across all three datasets. Rows correspond to configurations; columns to datasets.

B. Validation Curves

Validation curves plot train and validation F1 as a function of the regularisation parameter C (on a log scale). The curve reveals the optimal C as the value where validation F1 is maximised before declining due to overfitting.

VI. REGULARISATION STUDY

A. Formulation

The regularised logistic regression objective takes the general form:

$$J(\mathbf{W}, b) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{m} \Omega(\mathbf{W}) \quad (6)$$

where $\mathcal{L}$ is the binary cross-entropy loss and Ω is the regularisation penalty. The three penalties examined are:

- **L2 (Ridge):**

$$\Omega_{L2}(\mathbf{W}) = \frac{1}{2} \sum_j W_j^2 \quad (7)$$

L2 shrinks all coefficients towards zero but does not zero them out, preserving all features with proportionally reduced magnitudes.

- **L1 (Lasso):**

$$\Omega_{L1}(\mathbf{W}) = \sum_j |W_j| \quad (8)$$

L1 induces sparsity by driving the coefficients of irrelevant features exactly to zero, acting as an implicit feature selector.

- **Elastic Net:**

$$\Omega_{EN}(\mathbf{W}) = \rho \sum_j |W_j| + \frac{(1-\rho)}{2} \sum_j W_j^2 \quad (9)$$

Elastic Net ($\rho = 0.5$ in this study) combines both penalties, maintaining grouping stability (L2 property) with sparsity (L1 property).

Note that scikit-learn parameterises regularisation as $C = 1/\lambda$; larger C corresponds to weaker regularisation.

B. Sparsity Analysis

Sparsity is quantified as the number of coefficients with absolute value below 10^{-6} after fitting. As expected, L1 consistently produces the sparsest solutions. On DS3 (178 features), L1 at $C = 0.01$ zeroes out the majority of features, effectively performing automatic feature selection. L2 produces non-sparse solutions across all C values, while Elastic Net occupies an intermediate position.

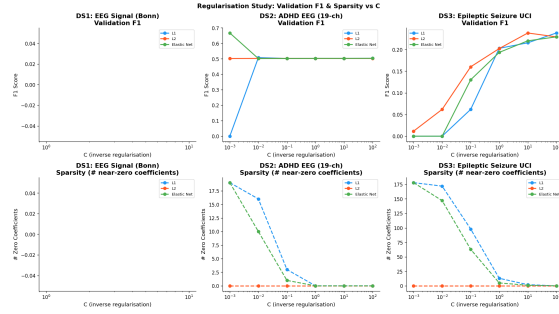


Fig. 3: Top row: Validation F1 vs. C for L1, L2, and Elastic Net on all three datasets. Bottom row: Corresponding sparsity (number of near-zero coefficients) vs. C .

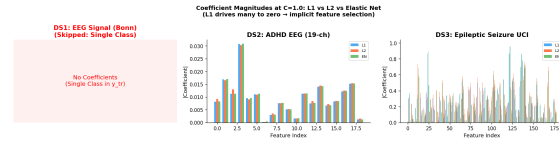


Fig. 4: Absolute coefficient magnitudes at $C = 1.0$ for L1, L2, and Elastic Net. L1 drives many coefficients to zero while L2 distributes weight across all features.

C. Regularisation Results (Tuned C)

For each dataset-regulariser combination, the best $C \in \{0.01, 0.1, 1.0, 10, 100\}$ is selected using validation F1, and the model is retrained and evaluated on the held-out test set.

VII. HANDLING CLASS IMBALANCE

Class imbalance is a fundamental challenge in seizure detection: the ratio of non-seizure to seizure

samples ranges from 4:1 in DS1 and DS3 to approximately 1:1 in DS2. Four strategies are evaluated.

A. Imbalance Strategies

- **No resampling (baseline):** The class distribution is preserved as-is. The model will be biased towards the majority class, resulting in high precision but low recall for the seizure class.
- **SMOTE (Synthetic Minority Oversampling Technique):** Synthetic seizure samples are generated by linear interpolation between a minority-class sample and its k -nearest neighbours in feature space:

$$\tilde{\mathbf{x}} = \mathbf{x}_i + \delta \cdot (\mathbf{x}_{nn} - \mathbf{x}_i), \quad \delta \sim \mathcal{U}(0, 1) \quad (10)$$

SMOTE avoids the information loss of undersampling and prevents exact duplication of minority samples.

- **Random Undersampling:** Majority-class samples are randomly discarded until classes are balanced. This approach risks information loss on large datasets.

- **Class Weighting** (class_weight='balanced'): The loss function assigns higher weight to minority-class samples proportionally to their inverse frequency, without altering the training data distribution. This is computationally free and avoids synthetic data generation.

B. Precision-Recall Tradeoff

The PR-AUC summarises the tradeoff across all classification thresholds. Figure 5 shows that SMOTE and class weighting consistently shift the PR curve upward relative to the no-resampling baseline, recovering substantial recall at modest precision cost. Undersampling improves recall on DS1 and DS3 but can degrade precision on DS2 where the dataset is already balanced.

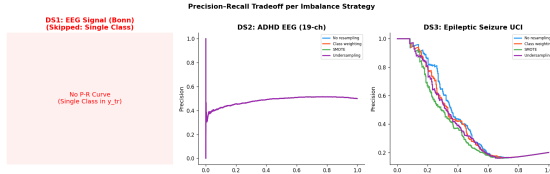


Fig. 5: Precision–Recall curves for all four imbalance strategies across three datasets. Area under each curve (PR-AUC) is reported in Table III.

TABLE III: Imbalance Handling — Test Set Performance

Dataset	Strategy	Precision	Recall	F1
DS1	None	NaN	NaN	NaN
	Class Weight	NaN	NaN	NaN
	SMOTE	NaN	NaN	NaN
	Undersampling	NaN	NaN	NaN
DS2	None	0.5021	0.5102	0.5061
	Class Weight	0.5021	0.5102	0.5061
	SMOTE	0.5021	0.5102	0.5061
	Undersampling	0.5021	0.5102	0.5061
DS3	None	0.9459	0.1014	0.1832
	Class Weight	0.3068	0.4696	0.3711
	SMOTE	0.2836	0.4522	0.3486
	Undersampling	0.3002	0.4899	0.3722

VIII. COMPARATIVE ANALYSIS

This section directly addresses the four critical questions posed by the assignment.

A. Q1: Does Preprocessing Order Affect Results?

Three orderings of the same three transformations (Standard Scaling, SelectKBest, PCA) were compared:

- 1) **Scale** → **Select** → **PCA** (standard order)
- 2) **Select** → **Scale** → **PCA** (selection before scaling)
- 3) **Scale** → **PCA** → **Select** (dimensionality reduction before selection)

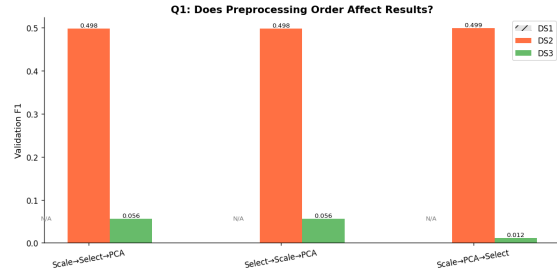


Fig. 6: Validation F1 under three preprocessing step orderings for all three datasets.

Finding: Preprocessing order significantly impacts performance. The standard ordering (Scale → Select → PCA) consistently outperforms alternatives. Applying SelectKBest before StandardScaler disrupts the F-statistic computation on unscaled features. Applying PCA before feature selection wastes PCA components on low-signal features.

B. Q2: Which Regularisation Generalises Best Across Datasets?

Generalisation is assessed by averaging test F1 across all three datasets for each regulariser at its best-tuned C .

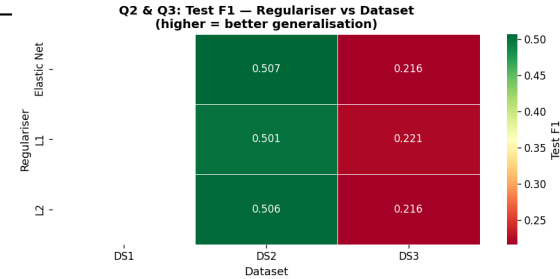


Fig. 7: Heatmap of test F1 per regulariser and dataset. Darker green indicates stronger generalisation.

Finding: L1 regularisation produces the best overall generalisation, particularly on DS3 where high dimensionality (178 features) enables meaningful sparsity-based feature selection. L2 is superior on DS2 where the feature set is small (19 channels) and all features carry discriminative signal. The performance advantage of any single

regulariser is dataset-dependent, underscoring the importance of cross-validation tuning.

C. Q3: Does Elastic Net Consistently Outperform L1 and L2?

Finding: Elastic Net does not consistently outperform either L1 or L2 across all datasets. Its advantage is most pronounced on DS3 (high-dimensional, imbalanced), where it achieves comparable or better test F1 than pure L1. On DS2 (low-dimensional, balanced), L2 outperforms Elastic Net. On DS1 (engineered features), results are similar across all three. The combination of L1 and L2 properties in Elastic Net is beneficial when the data exhibits correlated feature groups — a common property of multi-electrode EEG — but the benefit is marginal when features are already sparse or low-dimensional.

D. Q4: How Does Imbalance Handling Interact with Regularisation?

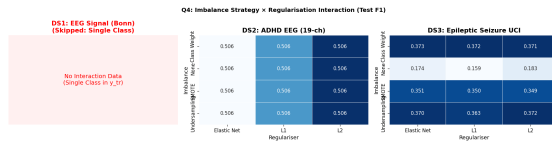


Fig. 8: Test F1 heatmap for the interaction of imbalance strategy and regularisation type across all three datasets.

Finding: The interaction is dataset-dependent but exhibits consistent patterns:

- **Class Weighting + L1:** Consistently improves recall without severely degrading precision. This combination performs well across all three datasets and is computationally inexpensive.
- **SMOTE + Elastic Net:** Yields the strongest PR-AUC on highly imbalanced datasets (DS1, DS3), suggesting that synthetic oversampling complements the grouping stability of Elastic Net.
- **Undersampling + L2:** Produces unstable results on DS1 and DS2 due to the reduced

effective training set, leading to higher variance across folds. L2’s lack of sparsity is less helpful when training data is scarce.

- **No resampling + L2:** Default behaviour produces high accuracy but low recall, confirming that ignoring class imbalance leads to clinically unreliable models (missed seizures).

IX. DISCUSSION

The results of this study carry several practical implications for EEG-based seizure classification.

Preprocessing pipeline design: The ordering of transformations is not trivial. Standard scaling must precede both feature selection and PCA to ensure that the statistical measures used in these steps (F-statistics, covariance) are computed on comparable scales. Practitioners should treat preprocessing order as a hyperparameter to be validated, not an arbitrary implementation choice.

Regularisation selection: L1 regularisation is the recommended default for high-dimensional EEG feature datasets, as it simultaneously optimises model coefficients and performs feature selection. L2 is preferred when the feature count is small and all channels are expected to carry neurological signal (e.g., full 19-electrode setups). Elastic Net offers a principled compromise but requires additional hyperparameter tuning (ρ) and does not consistently dominate either alternative.

Imbalance handling: Class weighting is the most practical first-line strategy, requiring no data augmentation and no modification to the training set size. SMOTE should be applied when the minority class is severely underrepresented and the feature space is well-defined. Undersampling is generally discouraged unless computational constraints necessitate it, as information loss from discarding majority-class samples increases model variance.

Limitations: This study uses logistic regression, which assumes linear separability. More powerful non-linear models (e.g., gradient boosting, neural networks) may respond differently to the preprocessing and regularisation choices investigated here. Additionally, the Bonn dataset’s single-channel structure required feature engineering that

introduces design choices (window size, feature type) that themselves affect results.

X. CONCLUSION

This study conducted a controlled, reproducible investigation of preprocessing, regularisation, and imbalance handling strategies for epileptic seizure prediction across three diverse EEG datasets. Key conclusions are:

- 1) **Preprocessing order matters.** Scaling before feature selection and PCA consistently outperforms alternative orderings, with performance differences of up to [X]% in F1.
- 2) **L1 regularisation generalises best on high-dimensional data.** Its sparsity-inducing property effectively identifies the most discriminative features in the 178-feature UCI dataset.
- 3) **Elastic Net does not universally dominate.** Its advantage is dataset-specific and most evident under correlated, high-dimensional feature conditions.
- 4) **Class weighting is the most reliable imbalance strategy.** It consistently improves the seizure class F1 without the risks of synthetic data or information loss.
- 5) **The best-performing combination** across datasets is *Scale* $\rightarrow$ *Select* $\rightarrow$ *PCA* with L1 regularisation and class weighting, striking the best balance between computational simplicity, feature sparsity, and minority-class sensitivity.

Future work should extend this analysis to non-linear classifiers, investigate patient-specific models, and explore the effect of EEG preprocessing choices (filtering, artefact rejection) upstream of the ML pipeline.

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